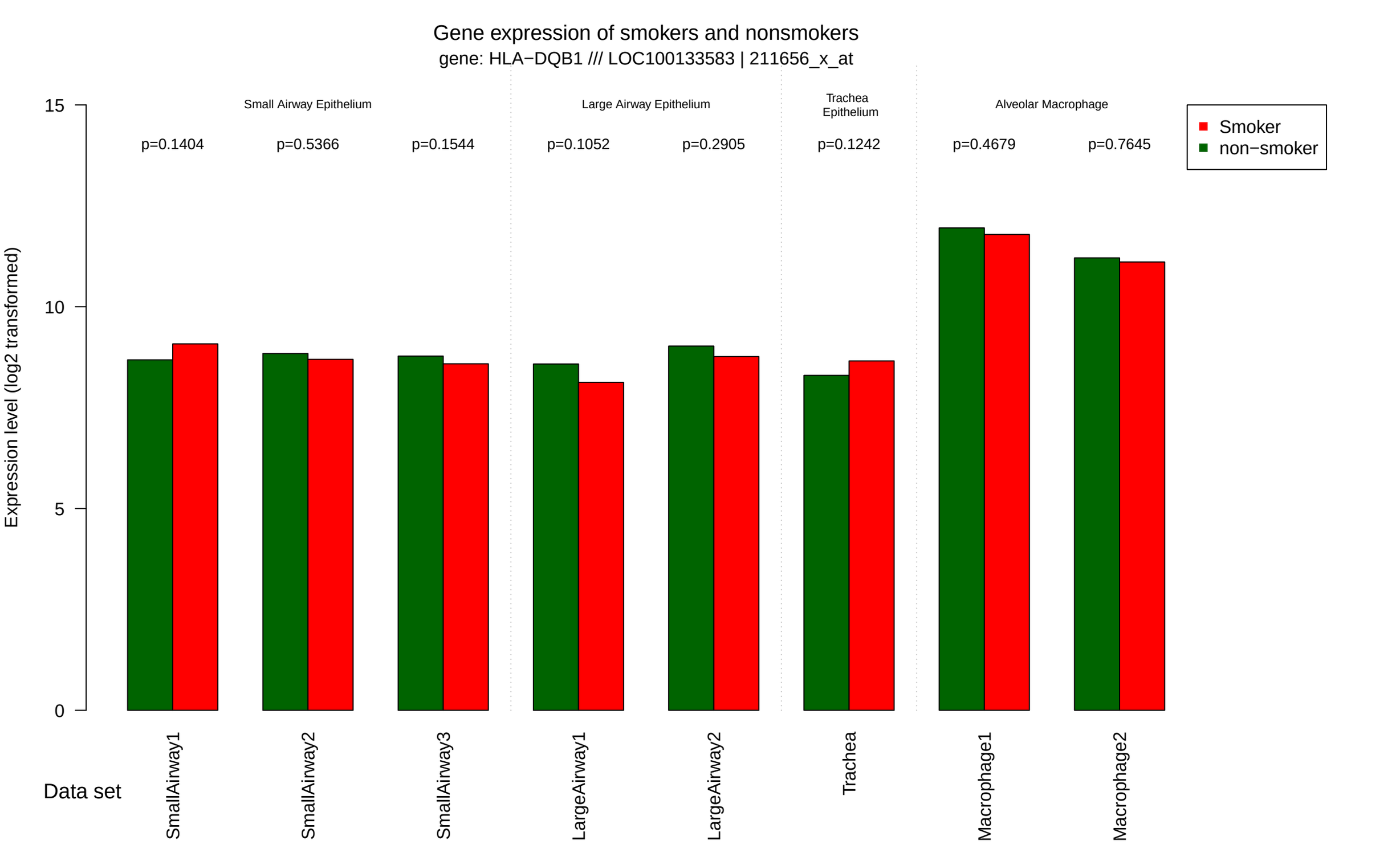
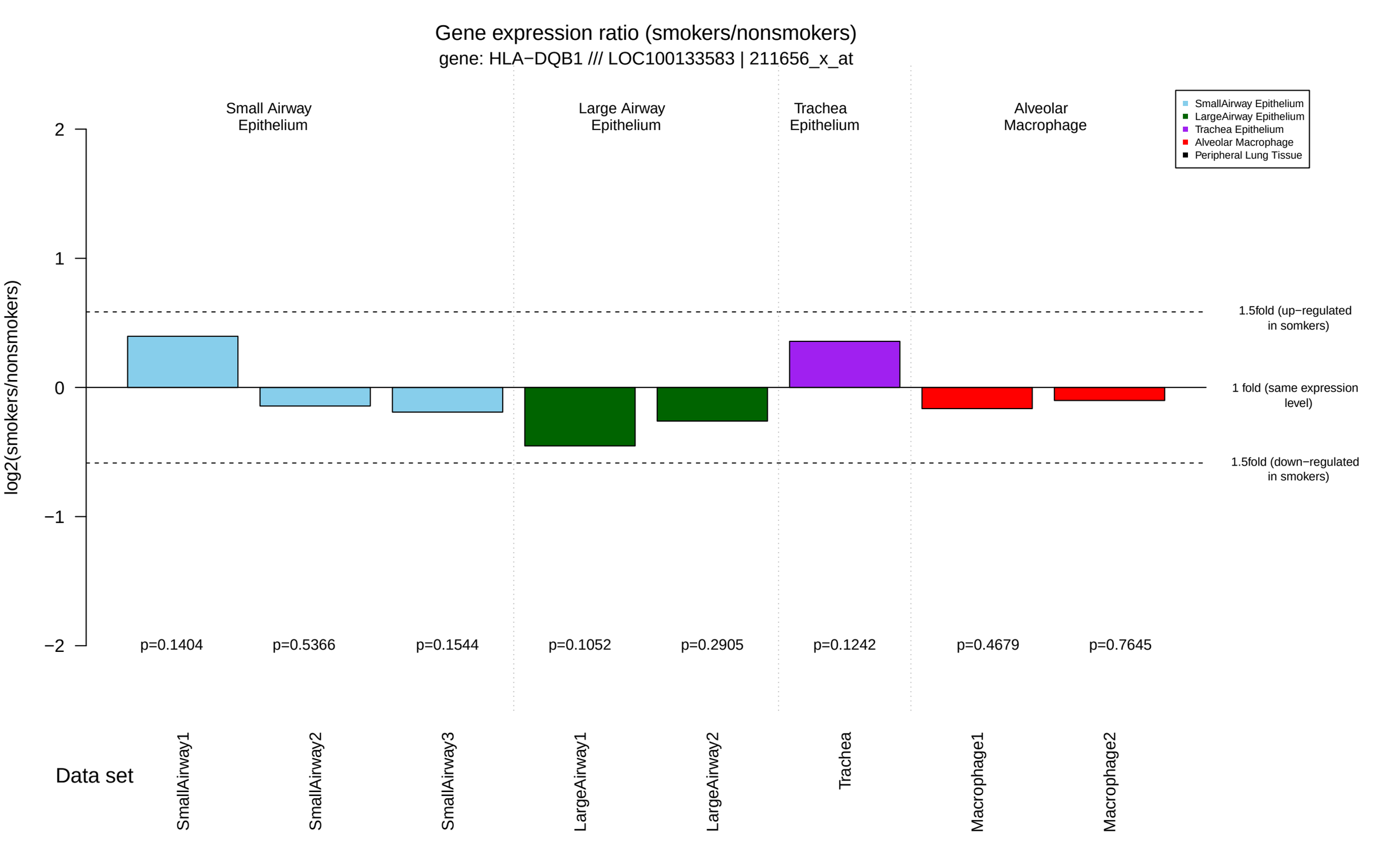
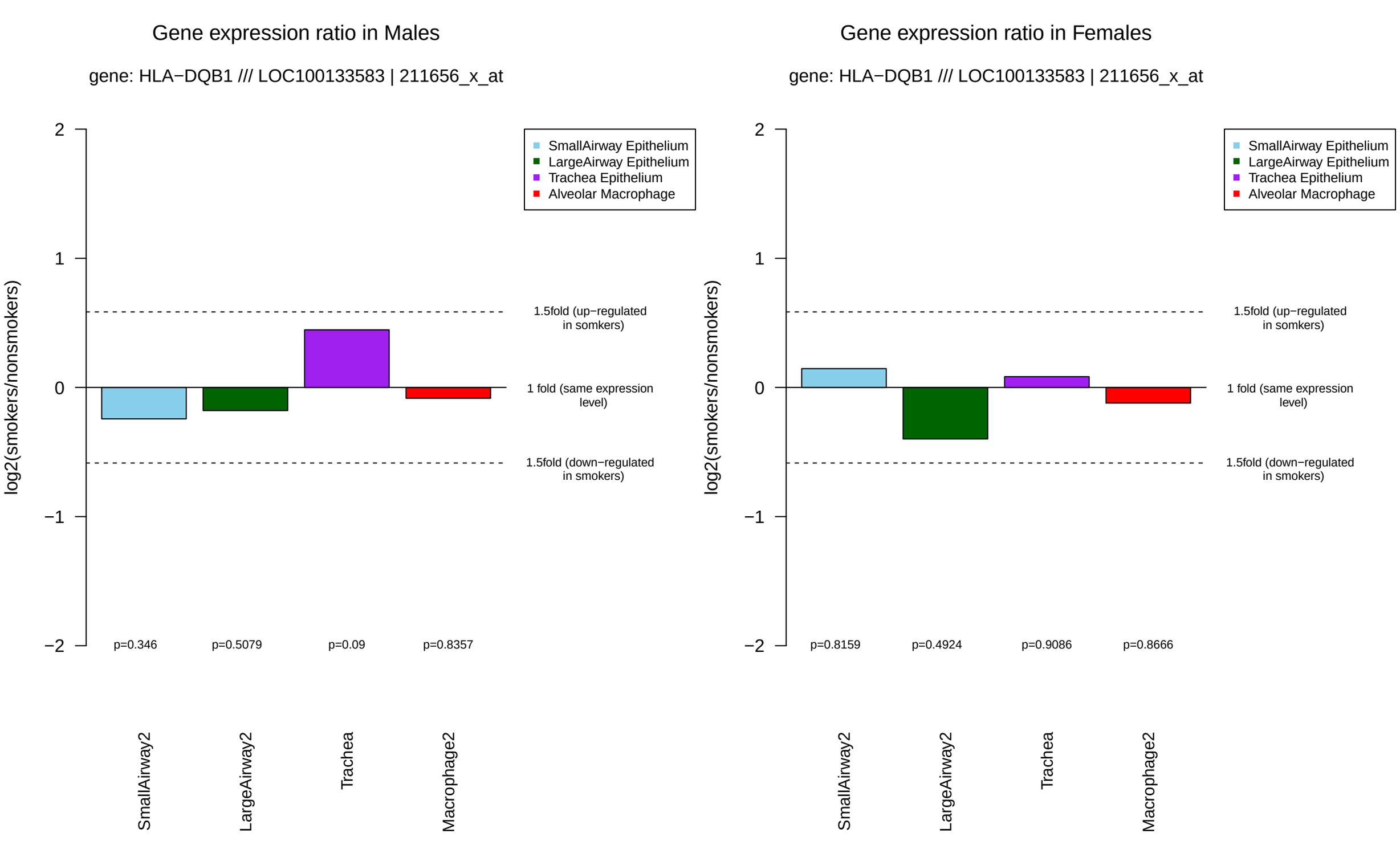
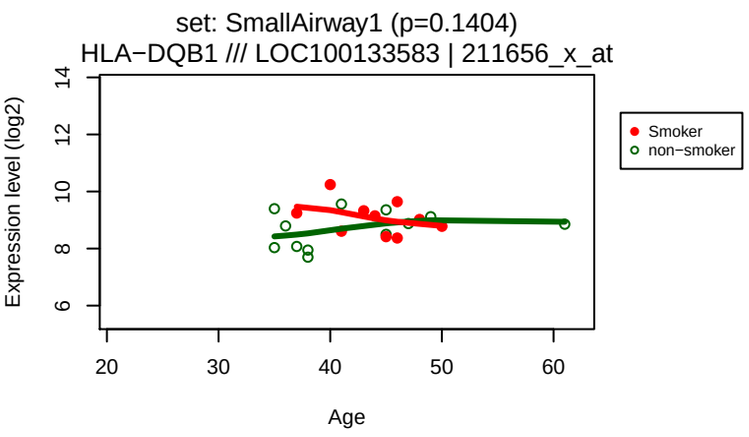
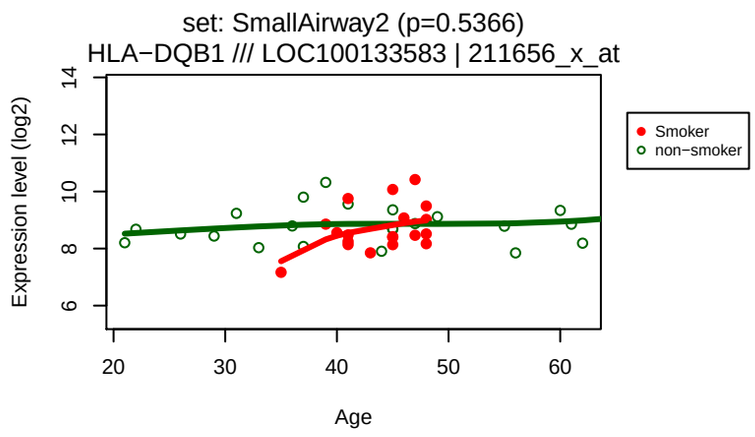
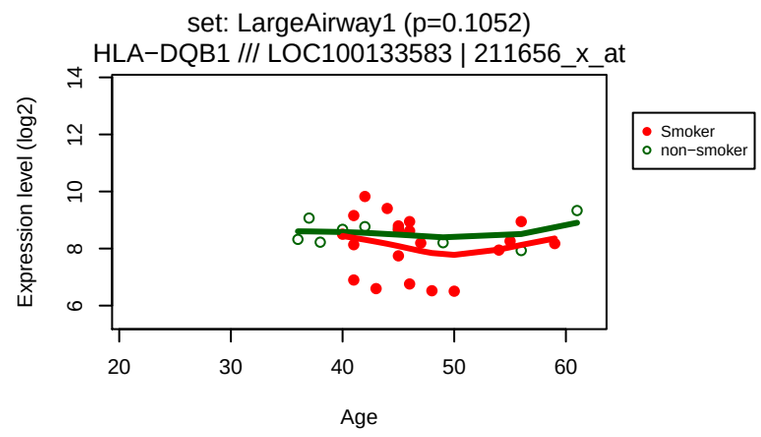

Gene expression of smokers and nonsmokers
gene: HLA−DQB1 /// LOC100133583 | 211656_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1404
p=0.5366
p=0.1544
p=0.1052
p=0.2905
p=0.1242
p=0.4679
p=0.7645
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

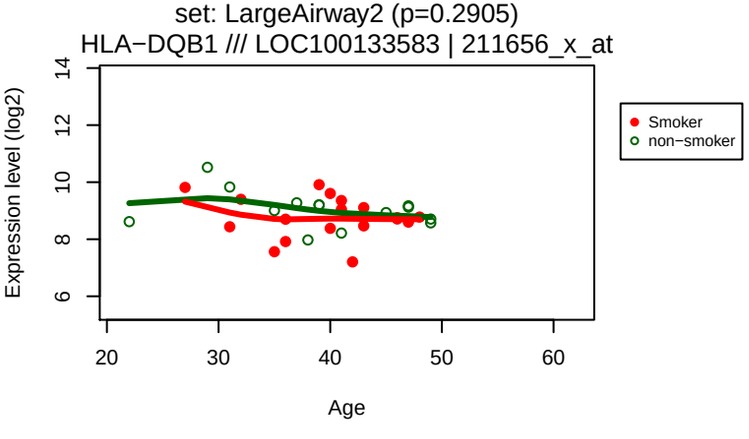

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DQB1 /// LOC100133583 | 211656_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1404
p=0.5366
p=0.1544
p=0.1052
p=0.2905
p=0.1242
p=0.4679
p=0.7645
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

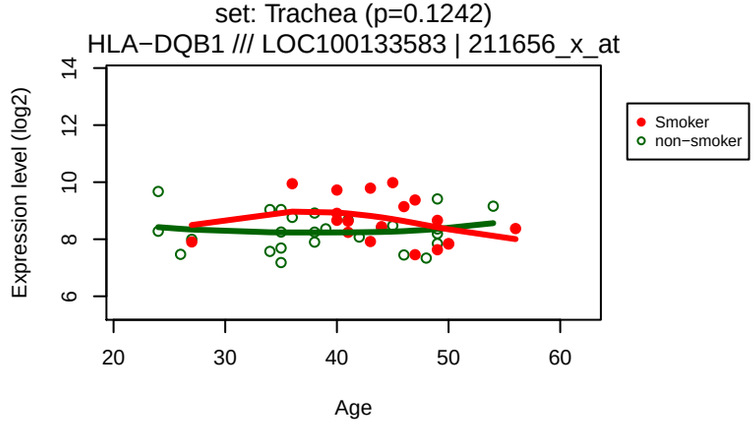

Gene expression ratio in Males
gene: HLA−DQB1 /// LOC100133583 | 211656_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.346
p=0.5079
p=0.09
p=0.8357
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DQB1 /// LOC100133583 | 211656_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8159
p=0.4924
p=0.9086
p=0.8666
SmallAirway2
LargeAirway2
Trachea
Macrophage2

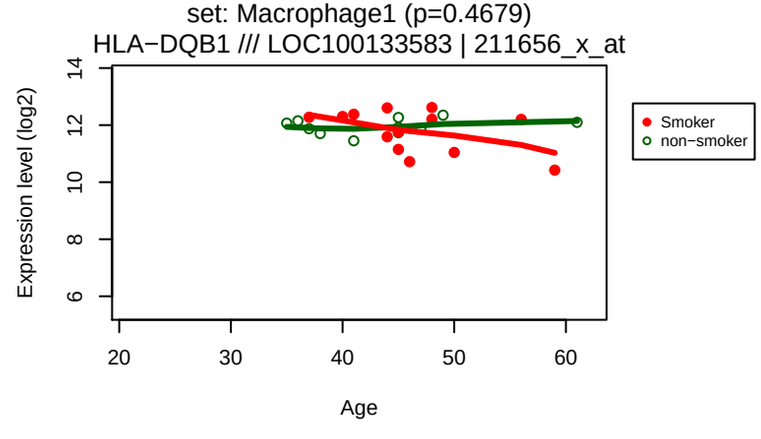

set: SmallAirway1 (p=0.1404)
HLA−DQB1 /// LOC100133583 | 211656_x_at
Expression level (log2)
Age
Smoker
non−smoker

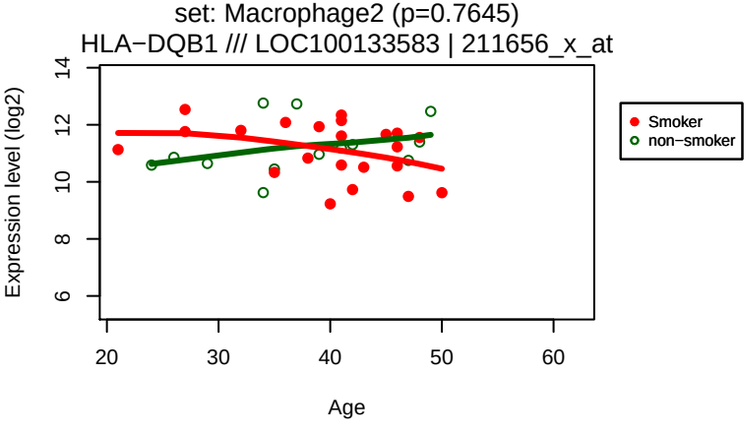

set: SmallAirway2 (p=0.5366)
HLA−DQB1 /// LOC100133583 | 211656_x_at
Expression level (log2)
Age
Smoker
non−smoker

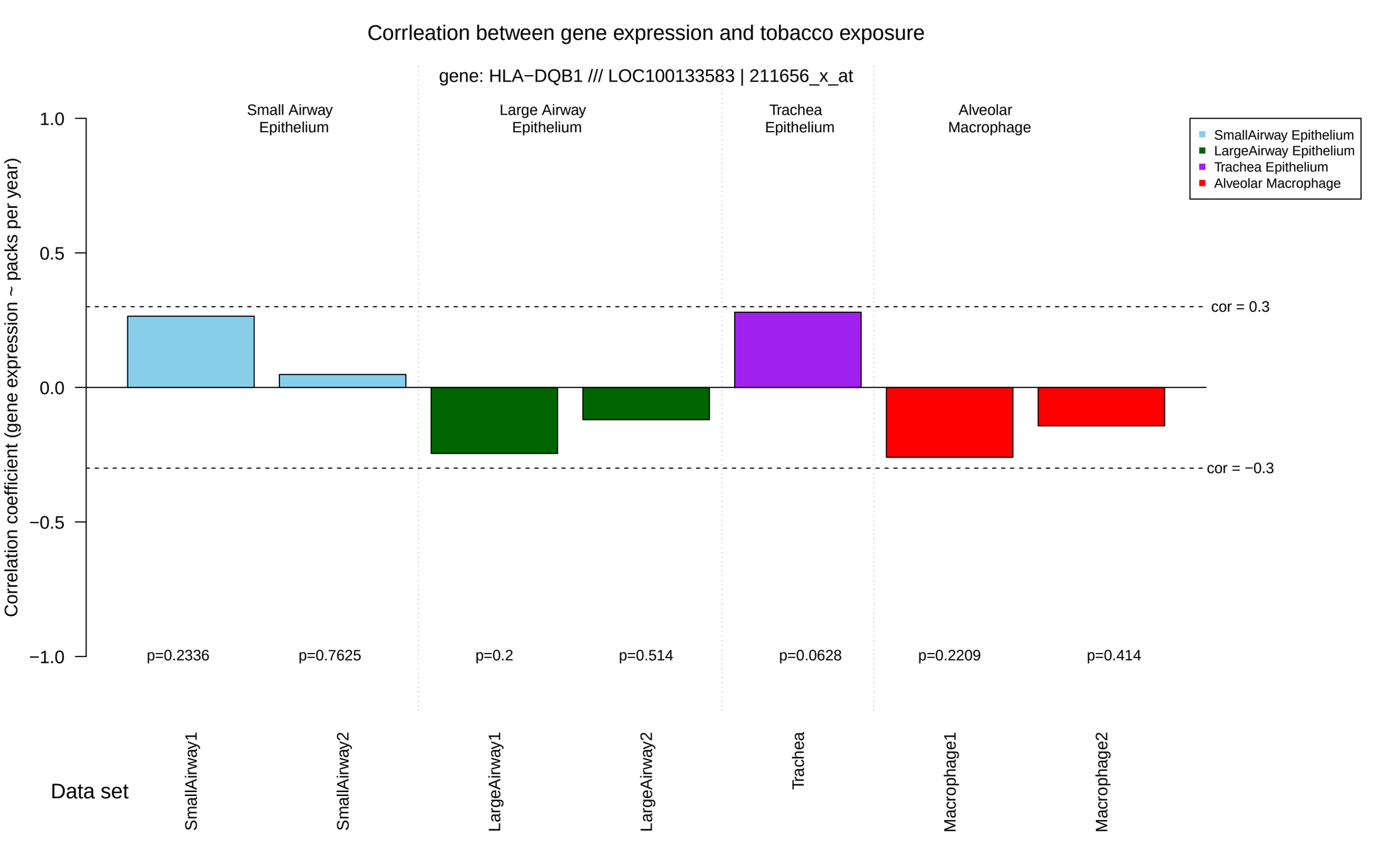

set: LargeAirway1 (p=0.1052)
HLA−DQB1 /// LOC100133583 | 211656_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2905)
HLA−DQB1 /// LOC100133583 | 211656_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1242)
HLA−DQB1 /// LOC100133583 | 211656_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4679)
HLA−DQB1 /// LOC100133583 | 211656_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7645)
HLA−DQB1 /// LOC100133583 | 211656_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DQB1 /// LOC100133583 | 211656_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2336
p=0.7625
p=0.2
p=0.514
p=0.0628
p=0.2209
p=0.414
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2